Zhiwei Zhou (周志威)

Email: zhou_zw@hust.edu.cn | zw.zhou.2000@gmail.com
Phone: +86 18290102978
Website: zhuizhou.github.io | Google Scholar
Address: Luoyu Road 1037, Wuhan, China

Education

Huazhong University of Science and Technology (HUST) | Wuhan, China

- **Ph.D. in Integrated Circuit Science and Engineering** | *Sept 2024 – Present*
  - Advisor: Prof. Yi Li (Odysseia Lab)
- **M.S. in Integrated Circuit Science and Engineering** | *Sept 2022 – Sept 2024*
  - Advisor: Prof. Yi Li (Odysseia Lab)
- **B.S. in Electronic Science and Technology** | *Sept 2018 – June 2022*

Research Interests

My research focuses on **Computing-in-Memory (CIM) Architecture and Circuit**, specifically:

- **Heterogeneous CIM Accelerators:** Designing hybrid RRAM-SRAM architectures tailored for Edge Large Language Models (LLMs) and Transformers.
- **Precision-Reconfigurable Computing:** Enabling unified Integer (INT) and Floating-Point (FP) arithmetic on analog CIM with arbitrary precision support.
- **Algorithm-Hardware Co-design:** Developing automated frameworks for quantization, mapping optimization, and variation-robust sparsity.

Publications

Selected Publications

- [**TCAS-I**] **Zhiwei Zhou†**, Jiancong Li†, Han Jia†, et al. "**ArPCIM: An Arbitrary-Precision Analog Computing-in-Memory Accelerator With Unified INT/FP Arithmetic**," *IEEE Transactions on Circuits and Systems I: Regular Papers*, 2025.
  - Proposed a reconfigurable analog CIM accelerator supporting precision from 1-bit to 47-bit.
  - Introduced a CIM-friendly local pre-alignment method with round truncation to minimize precision loss.
- [**ASP-DAC'26**] **Zhiwei Zhou**, Tong Hu, Han Bao, et al. "**OAH-CIM: Outlier-Aware Hybrid RRAM-SRAM CIM Accelerator with Variation-Robust Sparsity**," *Accepted by Asia and South Pacific Design Automation Conference*, 2026.
  - Developed an outlier-aware hybrid CIM architecture addressing the heavy-tailed activation distribution in Transformers.

    - Proposed Balanced Bit Sparsity to mitigate RRAM device variation noise during sparse computation.
- [**DAC'26 Under Review**] **Zhiwei Zhou**, Jiancong Li, et al. "**CIM-QMap: A Co-design Framework for RRAM-based Computing-In-Memory Architecture with Automatic Quantization and Mapping Optimization**".
    - Presented an automated framework with a unified metric simulator and a layer-wise pruning-guided DSE engine.
- [**ISCAS'26 Under Review**] **Zhiwei Zhou**, et al. "**Energy-Efficient Acceleration of Fourier-based Transformers on RRAM-CIM via Mixed-Precision and DFT Symmetry**."

Intellectual Property

Patents (Published)

1. **Yi Li, Zhiwei Zhou**, "A Universal In-Memory Matrix-Tensor Processor and Its Operation Method" , *Application No.: 2023105941388*.
2. **Yi Li, Zhiwei Zhou**, "A Joint Software-Hardware Simulation and Optimization System for Memristive Computing-in-Memory" , *Application No.: 2025110431717*.
3. **Yi Li, Zhiwei Zhou**, "A Universal Computing-in-Memory Simulation Method and System" , *Application No.: 2025110431736*.

Software Copyrights (Authorized)

1. **Yi Li, Zhiwei Zhou**, "Universal Memristive Computing-in-Memory Simulator Software V1.0" , *Registration No.: 2025SR1838416*.

Honors and Awards

- **National Second Prize** (Captain) | 8th China Postgraduate IC Innovation Competition | *2025.08*
- **National First Prize** (Member) | 7th China Postgraduate IC Innovation Competition | *2024.08*
- **National Third Prize** (Captain) | 20th China Postgraduate Mathematical Contest in Modeling | *2023.10*
- **National Second Prize** (Member) | 6th China Postgraduate IC Innovation Competition | *2023.08*
- **Outstanding Graduate** | Huazhong University of Science and Technology | *2022.06*

Project Experience

- **STI 2030—Major Projects (No. 2021ZD0201201)**
- **National Key R&D Plan of China (No. 2022YFB4500101)**

Technical Skills

- **EDA Tools:** Cadence (Virtuoso, Innovus), Synopsys (Design Compiler, PrimeTime), CACTI 7.0.
- **Languages:** Verilog HDL, Python (PyTorch), C/C++.
- **Frameworks:** Developer of *MemIntelli* (algorithm-hardware co-design simulation framework).